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RESEARCH ARTICLE

Oligomeric π-Extended Organic Mixed Ionic-Electronic Conductors for Intelligent Ion Sensing

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ABSTRACT

Realizing well-defined organic mixed ionic-electronic conductors (OMIECs) with more delocalized electronic structures remains challenging in intelligent and wearable bioelectronics. Here, we present a metal-free, step-economical π-extension strategy to construct biocompatible, monodisperse, fused all-acceptor oligomeric OMIECs (ONRs) for intelligent ion sensing. This approach established a clear relationship between molecular structure, microstructure, and device function. As one of the longest reported monodisperse OMIEC (45.7 Å), **4NR** delivers vertical organic electrochemical transistors (vOECTs) with high transconductance (7.42 mS), ultrafast switching (<1 ms), and excellent cycling stability (>90% retention), arising from its optimized microstructural balance that promotes ion accessibility while preserving long-range electronic connectivity. By leveraging voltage-dependent, reconfigurable n-type and anti-ambipolar behaviors, a waveform-based, non-invasive intelligent sweat-ion sensing platform integrated with machine-learning classification is established, enabling accurate discrimination of ionic concentration levels across physiologically relevant ranges, further highlighting the promise of precisely π-extended OMIECs for intelligent wearable bioelectronic sensing.

1 | Introduction

Organic electrochemical transistors (OECTs) have emerged as powerful platforms for wearable healthcare and neuromorphic computing due to their low power consumption, high transconductance, and excellent bio-interfacing capability [1–4]. Their unique bulk-doping mechanism enables efficient ion-electron transport within organic mixed ionic-electronic conductors (OMIECs), providing strong signal amplification and high sensitivity for detecting weak physiological stimuli [5–7]. By integrating adaptive calibration, on-device signal processing, and wireless communication, smart OECTs systems can perform real-time, closed-loop sensing and actuation, supporting next-generation AI-enabled, personalized healthcare [8, 9]. However,

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the advancement of such systems remains critically constrained by the lack of stable and high-performance n-type OECTs, which are fundamentally limited by the scarcity of efficient OMIECs. Although lowering the lowest unoccupied molecular orbital (LUMO) levels below −4.0 eV can suppress undesirable electrochemical side reactions, achieving both long-term operational stability and rapid response under aqueous and aerobic conditions continues to be a major challenge [10, 11]. Reliable n-type materials must thus combine deep energy levels, chemical robustness, rapid ionic/electronic coupling, and reproducible signal transduction to sustain algorithm-driven biosensing without degradation. However, establishing a controlled and reproducible balance between backbone electronic conductivity and side-chain-mediated packing relaxation is challenging in polydisperse polymer systems, owing to inherent batch-to-batch variations in molecular weight distribution and microstructural organization arising from polymer synthesis.

Monodisperse oligomeric OMIECs have recently emerged as a new class of precisely π-extended materials that bridge the gap between small molecules and polymers, offering molecular-level definition with suppressed defects, tunable crystallinity, and polymer-like flexibility without batch-to-batch variation and structural heterogeneity [12]. By judiciously extending conjugation length and optimizing molecular planarity, these systems achieve long-range charge delocalization, enhanced redox stability, and ordered packing conducive to efficient ionic-electronic transport [12–14]. Fused π-extended architectures with unique topologies and electronic structures further promote structural integrity and mitigate degradation during operational cycling in OECTs [15–17]. Nevertheless, their synthetic access remains limited by complex, multistep protocols requiring precise control over side-chain composition and conjugation motifs [18–20]. The development of concise, metal-free, and high-yielding synthetic strategies that reconcile molecular precision with functional performance is therefore essential to unlocking stable, high-mobility n-type OMIECs for intelligent, reconfigurable, and adaptive OECT-based bioelectronics [21, 22].

Here we report oligomeric π-extended OMIECs that enable OECTs with polarity-adaptive charge transport and voltage-modulated ion sensing capability (Figure 1a). A metal-free, step-economical π-extension strategy based on Knoevenagel condensation affords monodisperse, fused all-acceptor oligomers (ONRs) with well-defined backbones and controlled chain lengths. Among them, along with great compatibility, **4NR**, featuring a molecular length of 45.7 Å, represents one of the longest monodisperse π-extended OMIEC reported to date (Figure 1b). The π-extended, oligo(ethylene glycol) (OEG)-functionalized structurally programmable oligomers combine enhanced charge delocalization with excellent solution processability, while their moderately loosened molecular packing enables rapid switching with turn-on/off transients of 12.67/0.89 ms. These structurally well-defined monodisperse systems establish reliable structure-property correlations, allowing rational modulation of energy levels, molecular ordering, and ion-electron coupling for high-performance, stable n-type OECTs. Moreover, the voltage-dependent, reconfigurable n-type and anti-ambipolar transport behaviors highlight their potential in adaptive bioelectronic sensing. Leveraging these properties, a smart OECT-based sensing platform was further developed, where transient waveform features and peak current responses were analyzed via machine learning for rapid, non-invasive ion detection in body fluids. This study illustrates how a step-economical π-extension strategy bridges synthetic simplicity with data-guided, reconfigurable bioelectronic functionality, offering a versatile route toward intelligent and wearable healthcare systems.

2 | Results and Discussion

2.1 | Structural Design and DFT Modulation

Based on the previously reported **gNR** series and their oligomeric analogues, **4gNB**, bearing tetra-ethylene glycol side chains, was chosen as the core building block due to its excellent solubility, processability, and planarity, which favor efficient charge transport [15]. To further tune molecular packing and energy levels, 3-propyl rhodanine was employed as the terminal acceptor, following its proven performance in **gNR-Pr** [23]. Guided by this design, a series of fused all-acceptor oligomers (**2NR**, **3NR**, and **4NR**) with different conjugation lengths were synthesized via a step-economical π-extending condensation using tris(diethylamino)phosphine (TDEAP) as the reductant in toluene, followed by metal-free Knoevenagel end-capping to afford the final ONR derivatives (Figure 1c). The resulting oligomers, composed of two to four repeating units, were readily separated by column chromatography. Detailed synthetic procedures and the characterizations of all intermediates and ONRs are shown in Figures S1–S12. Their well-defined molecular structures and molecular masses were confirmed by high-temperature ^{1}H NMR, MALDI-TOF mass spectrometry, and elemental analysis (Figure S7–S12, Table S1). Both **2NR** and **3NR** exhibited sharp and well-resolved proton resonances, whereas **4NR** displayed broadened peaks indicative of polymer-like aggregation, yet still retained clearly assignable proton signals. Within experimental errors, the elemental analysis results of ONR derivatives were in good agreement with the calculated values. This step-economical and metal-free strategy eliminates residual transition-metal catalyst contamination and ensures high purity without batch effect, showing potential for future development toward biocompatible bioelectronic materials [15, 24, 25]. Moreover, the monodisperse π-extended OMIECs overcome the inherent batch-to-batch variation and suppress defects, which are inherent challenges in conjugated polymers, thereby enabling reproducible synthesis and precise establishment of structure-property relationships [26]. All ONRs show excellent solubility in common solvents such as chloroform and hexafluoroisopropanol (HFIP), facilitating solution-based device fabrication [27]. Together, these results establish a reliable synthetic foundation for subsequent investigations of the structure, microstructure, and OECT performance of the π-extended n-type OMIECs.

Simulations were performed to elucidate the geometrical, electronic, and molecular packing characteristics of the ONR derivatives. As shown in Figure S13, the density functional theory (DFT) results show that the rhodanine and **NB** units maintained good planarity, while dihedral angles between adjacent **NB** units are approximately 19°, arising from steric hindrance. This moderate torsion leads to relatively loose stacking, providing additional free volume and dynamic pathways that facilitate efficient ion penetration and injection [28, 29]. The LUMO wavefunctions

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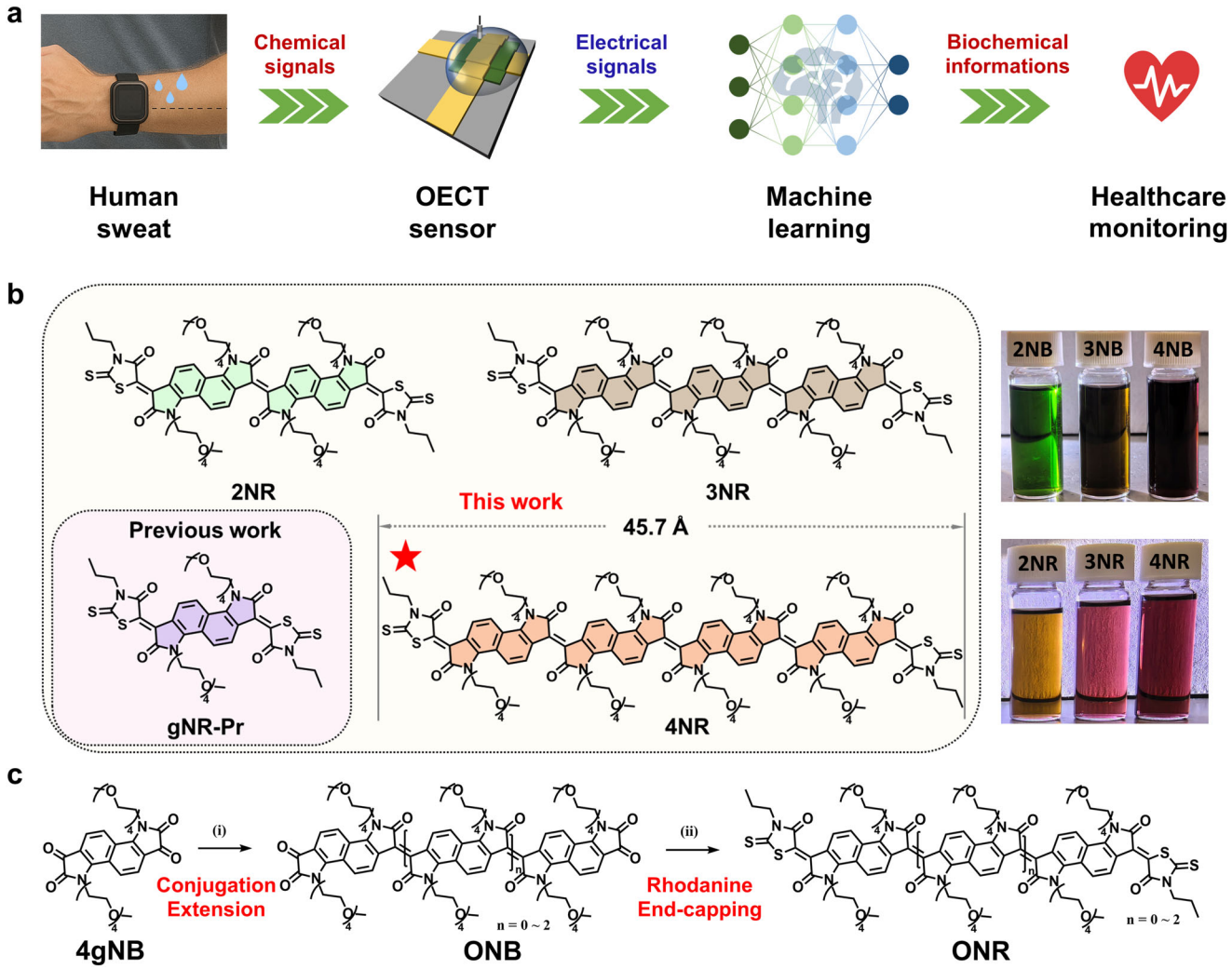


FIGURE 1 | (a) Diagram of human sweat detection based on ONRs-based OECT devices; (b) Molecular structures and images of ONBs and ONRs dissolved in chloroform; (c) Synthetic routes of ONRs.

(Figure S14) were well delocalized along the conjugated backbone, indicating strong potential for long-range electron transport [30]. To further probe molecular packing and ion accessibility, molecular dynamics simulations were performed on ONRs bearing OEG side chains. Initial structures were submitted to energy minimization and with conjugate gradient algorithm. The minimized systems were then equilibrated under an NVT ensemble at 300 K for 5 ns employing the velocity-rescale thermostat. Subsequently, a production simulation was carried out in the NPT ensemble at 300 K and 1 bar, with Berendsen temperature and pressure coupling. Representative configurations obtained from the simulations are shown in Figure S15. Radial distribution function (RDF) analyses were performed using the final trajectory of the NPT-50 ns simulation to evaluate the short-range intermolecular correlations and ion coordination in the three systems (Figure 2) [31]. **2NR** displayed the strongest backbone correlation peak, while **3NR** and **4NR** showed progressively weaker interactions (Figure 2a), consistent with looser molecular packing in longer oligomers. Conversely, RDFs between Na^+/Cl^- ions and the conjugated backbones revealed enhanced ion coordination probabilities for **4NR**, implying that looser packing promotes ion accessibility (Figure 2b) [32]. Notably, Cl^- ions exhibited weaker interactions than Na^+ (Figure 2c), consistent with the n-type character of ONRs. Overall, the combination of controlled steric modulation, delocalized frontier orbitals, and packing-dependent ion transport underscores the suitability of these precisely π-extended oligomers as high-performance n-type OMIECs.

2.2 | Optical, Electrical, and Electrochemical Characterizations

To analyze the optical properties of the oligomeric OMIECs, Visible-Near Infrared (vis–NIR) absorptions in both solutions and thin films were conducted (Figure 2d,e). In chloroform, the π–π^* absorption peaks appeared around 510 nm, while in thin films, a slight bathochromic-shift was observed with increasing elongation length (501 nm for **2NR**, 504 nm for **3NR**, and 508 nm for **4NR**), indicating similar aggregation behaviors across the series. Notably, intramolecular charge transfer (ICT) absorption displayed pronounced red shifts, which we attributed to the asymmetric electron affinities of rhodanine and **NB** fragments. The ICT peaks in films were located at 792, 818, and 846 nm for

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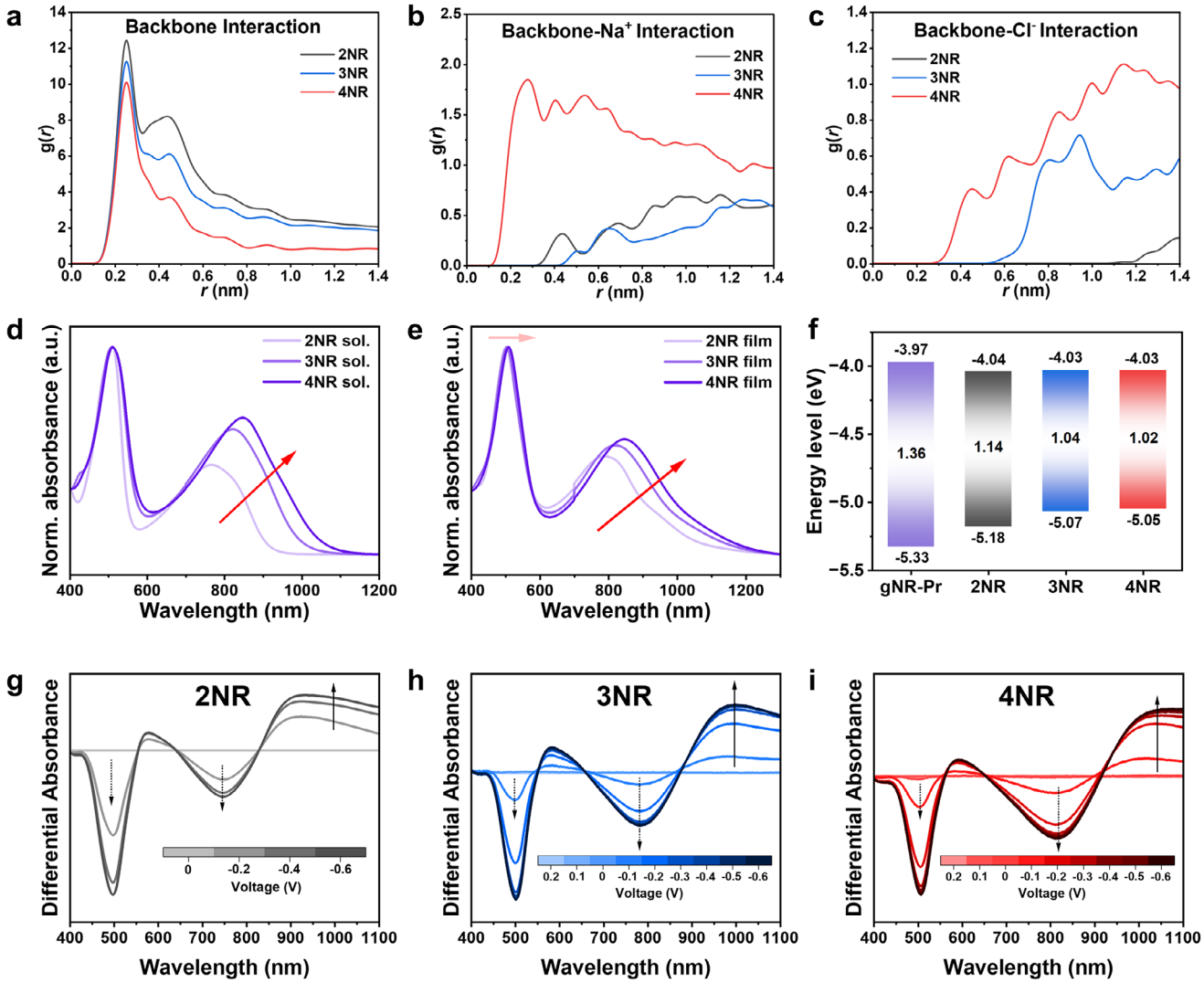


FIGURE 2 | RDF plots of molecular backbone and ion proximity on (a) backbone interaction, (b) backbone-Na^+ interaction, and (c) backbone-Cl^- interaction calculated from the final trajectory of the NPT-50 ns; Normalized Vis-NIR absorption spectra of ONRs in (d) chloroform solution and (e) thin films; (f) Molecular orbital energy levels of ONRs and **gNR-Pr**; Difference spectroelectrochemical spectra of (g) **2NR**, (h) **3NR**, and (i) **4NR** in 0.1 M aqueous sodium chloride electrolyte.

2NR, **3NR**, and **4NR**, respectively, closely matching trends in solution (765, 821, and 847 nm). This systematic shift confirms effective conjugation extension, facilitating charge transport even with suboptimal molecular planarity. The monomer **gNR-Pr** showed the shortest ICT absorption onset (λ_{onset}^{abs}) at 912 nm, while **2NR**, **3NR**, and **4NR** exhibited progressively red-shifted onsets to 1085, 1190, and 1214 nm, extending optical activity into the NIR-II region [23]. The reduced extent of shift with chain elongation implies saturation of the effective conjugation [33]. From these onsets, the optical band gaps (E_g^{opt}) were determined as 1.14, 1.04, and 1.02 eV for **2NR**, **3NR**, and **4NR**, respectively. The narrower gap of **4NR** reflects enhanced π-conjugation, which is typically associated with more delocalized charge carriers and can facilitate improved charge transport [34].

Electrochemical analysis via reductive cyclic voltammetry (CV) in chloroform (Figure S16) revealed deep-lying LUMO energies below −4.0 eV for all oligomers (−4.04, −4.03, and −4.03 eV for **2NR**, **3NR**, and **4NR**), supporting ambient stability of n-type operation [35]. The deep LUMO energies are attributed to electron delocalization across the extended π-backbone and the electron-withdrawing nature of the **NB** and rhodanine fragments, which stabilizes the added electrons in the reduction process. To simulate actual electrochemical operating states, the first reduction potentials (E_{1r}) in aqueous NaCl (0.1 M) shifted positively from −0.05 V for **2NR** to 0 V for **4NR** (Figure S17), indicating lower electron injection barriers and higher electron affinity with longer conjugation [36]. The energy levels of highest occupied molecular orbitals (HOMO) were obtained by subtracting E_g^{opt} from LUMO levels (Figure 2f). The HOMO energy levels exhibited an upward trend with conjugation extension and nearly approached saturation at longer lengths. This saturation behavior stems from the fact that the π-electron delocalization approaches the limit, making the incremental contribution of additional repeating units to the HOMO level negligible [37].

Spectroelectrochemistry in the aqueous electrolyte was used to probe redox-driven doping processes relevant to OECT operation.

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While **2NR** suffered from film stripping under bias, likely due to lower molecular weight and poor adhesion, both **3NR** and **4NR** exhibited stable electrochemical behavior over potentials from 0.2 to −0.6 V (Figure 2g–i). Under negative bias, all ONR derivatives showed strong suppression of the π–π^* and ICT bands, accompanied by the emergence of broad NIR-II absorptions characteristic of polaronic formation. These spectral changes indicate efficient charge injection and stable polaron formation [38]. The presence of well-defined isosbestic points at 828, 876, and 918 nm for **2NR**, **3NR**, and **4NR**, respectively, highlights the clean interconversion between neutral and reduced species. To intuitively analyze the doping characteristics of the ONR derivatives, the normalized absorbance changes were subtracted by the initial values at 0.2 V for **3NR** and **4NR** (0 V for **2NR**), thereby revealing the evolution of absorption peaks under applied potentials. Difference absorbance analysis (Figure S18) further revealed pronounced transitions in the π–π^*, ICT, and polaron bands for **3NR** and **4NR** compared to **2NR**, underscoring enhanced ion uptake and improved doping efficiency in the longer oligomers.

TD-DFT calculations provide mechanistic support for the excited-state characteristics of polaron formation (Table S3). The calculated excited state energies are in good agreement with the experimental red-shifting trend from **2NR** to **4NR** in the neutral state. To investigate the effect of Na^+ on the molecular geometry and optical absorption properties, anionic systems with net charges of −1, −2, and −3 (ONR^-: Na^+, ONR^{2-}: $2Na^+$, ONR^{3-}: $3Na^+$) were modeled by adding one to three sodium ions, respectively (Figure S19). Structural comparisons also revealed increased molecular twisting upon Na^+ coordination, with terminal dihedral angles rising from nearly 0° to ∼21°, suggesting that ion binding occurs near the terminal rhodanine moieties. Moreover, open-shell anionic states (ONR^- and ONR^{3-}) exhibit more red-shifted, lower-energy transitions than the closed-shell dianion (ONR^{2-}) because the presence of an unpaired electron generates polaronic mid-gap orbitals that significantly narrow the excitation energy gap. In contrast, the ONR^{2-} species forms a paired bi-polaron, where stronger electron-electron repulsion and reduced charge delocalization widen the HOMO-LUMO gap, leading to higher-energy transitions. Additionally, the radical anions undergo greater geometric relaxation upon reduction, which further stabilizes their excited states and contributes to the pronounced red-shift observed for the −1 and −3 charge states (Table S3).

2.3 | Micro-structural Properties

To elucidate the influence of conjugation length on microstructure and electrical properties, grazing-incidence wide-angle X-ray scattering (GIWAXS) measurements were performed on thin films of the ONR derivatives. Both pristine films and those blended with long-chain double-end trifluoromethyl diazirine (DtFDA) and cinnamate-cellulose polymer (Cin-Cell) were examined [39]. The blend system, suffixed with DC, incorporates DtFDA as a photo-crosslinker to enhance structural robustness, and Cin-Cell as a mechanical support for improved film integrity and modulated ion transport (vide infra). Comparison of the GIWAXS patterns (Figure 3a–f and Figures S20 and S21) reveals that crosslinking has a negligible impact on molecular packing. All three materials predominantly adopt an edge-on orientation, but differ in crystallinity. For **2NR**, the shortest conjugation length leads to more localized π-electrons, reduced conformational flexibility, and lower steric hindrance, collectively favoring tighter and more ordered molecular packing [40]. Accordingly, **2NR** exhibits additional (300) diffraction peak at $q_z = 0.326$ Å^{-1}, indicative of enhanced crystallinity. Meanwhile, **2NR** exhibits two distinct (010) peaks located at $q_{xy} = 1.5$–2.0 Å^{-1}, indicating heterogeneous π–π stacking environments rather than a well-defined ordered packing, corresponding to distances of 3.36 and 3.67 Å, respectively. In contrast, **3NR** and **4NR** display only (100), (200), and sole (010) reflections, suggesting reduced intermolecular order. This reduction in apparent crystallinity correlates with increased backbone length and steric interactions among side chains, which may introduce local torsional distortions and partially disrupt π–π stacking coherence [41].

The π–π stacking distances of 3.50 and 3.52 Å for **3NR** and **4NR,** respectively, also indicate progressively reduced crystallinity as the π-conjugation extends. The lamellar spacing increases progressively with backbone extension, with first-order lamellar diffraction peaks at 17.60, 19.70, and 21.23 Å for **2NR**, **3NR**, and **4NR**, respectively. Notably, the expanded lamellar spacing observed for **4NR** is consistent with reduced side-chain packing coherence, which may increase the availability of ion-accessible free volume within the film [42]. To intuitively compare the crystallinity differences among the **ONR** derivatives, correlation lengths (CL) were calculated (Tables S4 and S5). **2NR** possessed the highest CL values along both the (100) IP direction and (010) OOP direction, at 146.88 and 43.77 Å, respectively, indicating the highest degree of long-range structural correlation. Interestingly, **4NR** maintained a (010) π–π packing CL of 42.94 Å, comparable to that of **2NR**, but displayed a markedly reduced lamellar CL along the (100) IP direction, reflecting that local backbone π-stacking correlations were preserved, whereas side-chain-dominated lamellar ordering became weakly correlated. This indicates that efficient short-range charge transport coexists with relaxed long-range packing, and that enhanced ion accessibility is not achieved at the expense of backbone ordering upon extension of the π-conjugation. Such diminished side-chain coherence can benefit ion permeation in OECTs and other OMIEC-based devices by introducing interconnected free-volume pathways that facilitate bulk ion transport [43]. A schematic illustration of the molecular packing parameters is shown in Figure 3g–i. Although reduced long-range crystallinity may hinder charge transport, the extended conjugation of **4NR** enhances intramolecular charge transfer, partially compensating for this drawback. Moreover, the more relaxed molecular packing improves ion infiltration and mobility, rendering **4NR** a promising high-performance n-type OMIEC [44].

Atomic force microscopy (AFM) further validated the stacking behavior inferred from GIWAXS (Figure S22–S25). All three films exhibited smooth, grain-like morphologies. **4NR** showed the roughest surface with a root-mean-square (RMS) roughness of 12.6 nm, compared with 4.3 nm for **2NR** and 8.9 nm for **3NR**. This trend correlates with reduced long-range lamellar order while maintaining continuous film coverage without macroscopic defects, further confirming the superior crystallinity of **2NR** and the slightly more disordered packing of **3NR** and **4NR**. These structural characterizations indicate appreciable changes in

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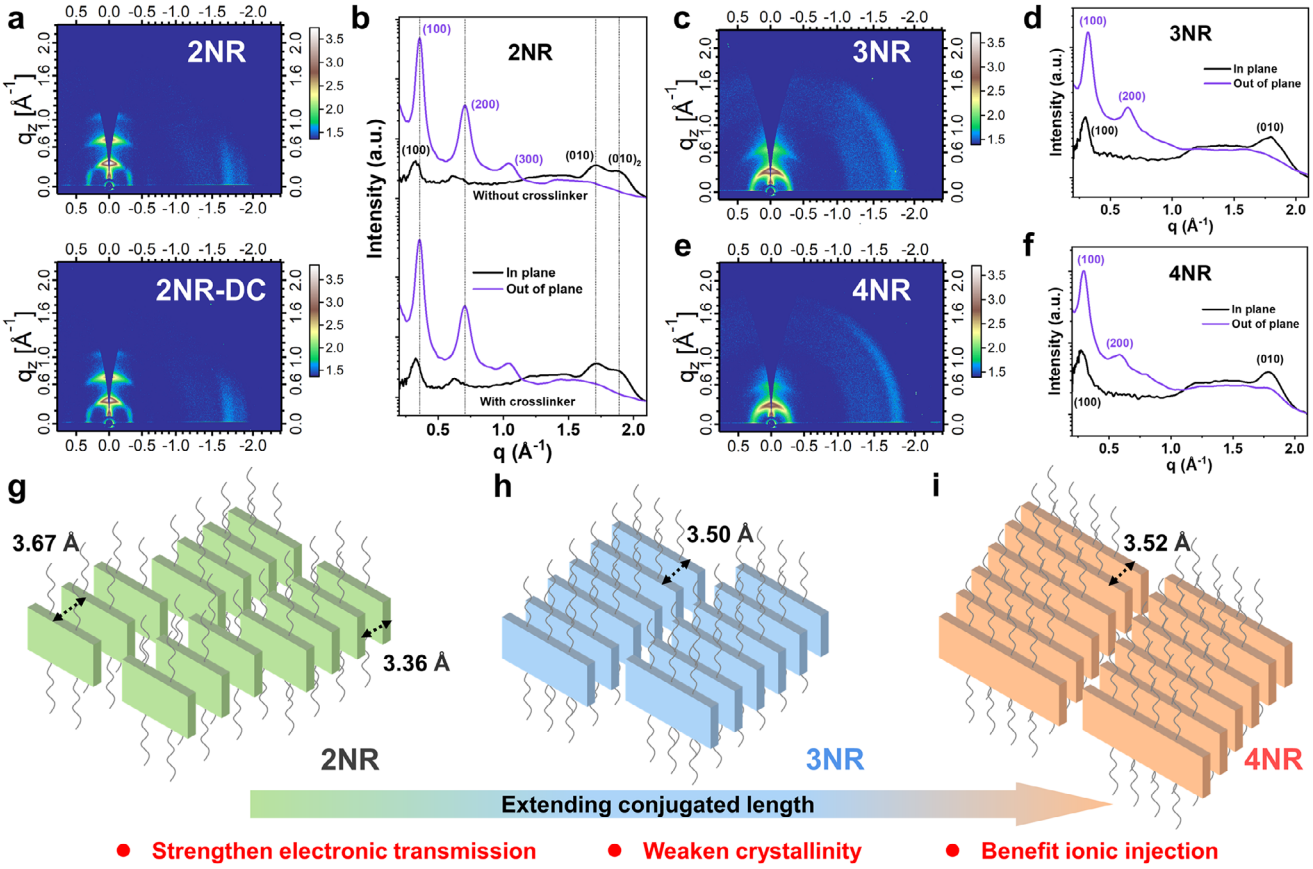


FIGURE 3 | GIWAXS characterization and stacking diagram. (a) 2D GIWAXS images of thin films of **2NR** and **2NR-DC**; (b) 1D GIWAXS charts of **2NR** and **2NR-DC** extracted by integration along q_{xy} (IP) and q_z (OOP) direction; 2D GIWAXS images of thin films of (c) **3NR** and (e) **4NR**; 1D GIWAXS charts of (d) **3NR** and (f) **4NR** extracted by integration along q_{xy} (IP) and q_z (OOP) direction; Diagram of molecular stacking pattern of (g) **2NR** and (h) **3NR**, and (i) **4NR**.

molecular packing with increasing conjugation length, expected to influence ion accessibility and electronic transport in OECTs.

2.4 | Organic Electrochemical Transistor Characteristics

Guided by the promising electrochemical performance of **4NR**, vertical OECTs (vOECT) based on ONR derivatives were fabricated and evaluated using the sandwich-type architecture illustrated in Figure 4a. To enable a meaningful comparison among the oligomers, all devices were fabricated using identical channel dimensions ($W = d = 40$ μm). DtFDA and Cin-Cell were incorporated into the active-layer formulation [39]. 10 mM phosphate-buffered saline (PBS) was employed as the electrolyte owing to its biocompatibility, pH stability, and suitable ionic conductivity, closely mimicking physiological conditions [45].

As shown in Figure 4b and Figure S26, all three devices demonstrate n-type characteristics with strong signal amplification, achieving high on/off current ratios (I_{on}/I_{off}) of ~10^6 (all vOECT parameters are summarized in Table S6). I_{on} increases markedly with the conjugation length, reaching 1.85 mA for **4NR**, compared with 0.35 and 0.81 mA for **2NR** and **3NR**, respectively (Figure 4c). The enhanced current output of **4NR** arises from synergistic electronic and ionic contributions. Although **4NR** exhibited reduced crystallinity relative to **2NR**, its moderately loosened packing improved ion accessibility without significantly impairing electronic percolation. A higher voltage at peak current of 0.55 V was recorded for **4NR,** compared with 0.45 V for **2NR** and 0.47 V for **3NR**, respectively. This shift suggests a greater degree of ionic penetration is required to fully dope the bulk channel, consistent with its enlarged lamellar spacing [46].

Although **4NR** exhibits the loosest molecular packing among the series, it exhibited a slightly higher threshold voltage (V_{th}) at −0.115 V, consistent with CV and electrochemical measurements, reflecting a higher initial barrier for n-type doping [47]. This indicates that the onset of ion penetration does not directly translate into effective channel formation. The extended π-conjugation in **4NR** promotes charge delocalization over longer molecular length scales, mitigating the impact of localized doping at low gate biases. The ultra-low V_{th} of ONR derivatives contributes to low-power transistor devices. All devices demonstrated outstanding operational stability, retaining over 90% of their initial current after 300 electrochemical cycles (**2NR**: 93.9%; **3NR**: 92.4%; **4NR**: 92.9%; Figure S27), indicating excellent stability in n-type small-molecule OECT devices. Extended testing further confirmed the durability of **4NR**, maintaining 83% of its initial current after 10 000 cycles (Figure 4d) among the best reported n-type OMIECs. This exceptional stability is attributed to the relatively low LUMO energy levels and the restricted ionic swelling afforded

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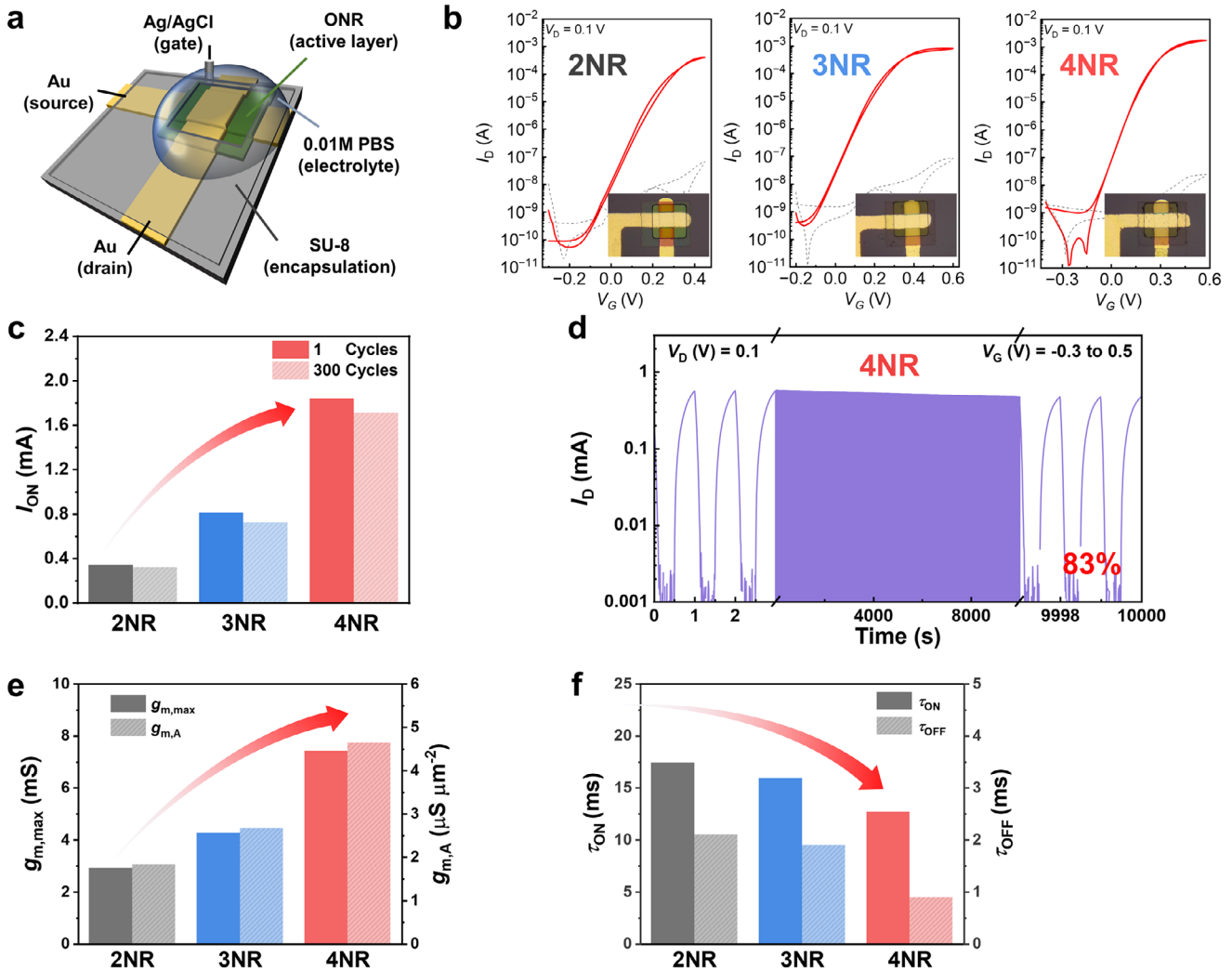


FIGURE 4 | Vertical organic electrochemical transistor behaviors. (a) Diagram of ONRs based vOECTs; (b) Transfer curves and optical microscope images of ONRs based vOECT devices, and **2NR** were tested to only 0.4 V for the emergence of anti-ambipolar characteristic; (c) On currents of ONRs based vOECT devices over 1 and 300 cycles; (d) Operational stabilities of **4NR** over 10 000 cycles of continuous electrochemical biases with an alternating gate potential ranging from −0.3 to 0.5 V and a pulse duration of 0.5 s for vOECT devices biased at 0.1 V in 0.01 M PBS; (e) Maximum transconductances of ONRs; (f) On and OFF time constant of vOECT based on ONRs.

by compact small-molecule packing and encapsulation (see Table S6). The voltage control over current ($g_m = \frac{\partial I_D}{\partial V_G}$, $g_{m,A} = \frac{g_m}{W \cdot d}$), for all ONR derivatives is summarized in Figure 4e. Among them, **4NR** exhibited the highest signal amplification capability, with a maximum transconductance ($g_{m,max}$) of 7.42 mS, surpassing **3NR** (4.26 mS) and **2NR** (2.91 mS), indicative of excellent ionic-electronic coupling. This superior performance correlates with the synergy effect of high electron mobility, enabled by the extended conjugation length, and efficient ion permeation, facilitated by moderate crystallinity. A balanced molecular ordering also provides sufficient ion channels, supporting rapid and reversible ionic uptake and release that accelerates doping/de-doping kinetics [48].

4NR exhibited the fastest transient response, with turn-on (τ_{ON}) and turn-off (τ_{OFF}) times of 12.67 and 0.89 ms, respectively, compared with 15.90/1.90 ms for **3NR** and 17.42/2.10 ms for **2NR** (Figure 4f and Figure S28). These results indicate a very fast response, primarily enabled by the vertical transistor architecture, in which ion transport occurs through a nanometer-scale thickness rather than laterally across micrometer channels, minimizing ion diffusion distance and expediting capacitive charging [49]. Such rapid switching dynamics enable ONR-based devices to monitor ionic or biochemical signals in real time with high temporal resolution. Furthermore, the monodisperse oligomeric structure of ONRs ensures well-defined backbones and controlled π-stacking, enhancing charge mobility and facilitating efficient ionic-electronic coupling. The fast doping/de-doping behavior also contributes to operational stability by reducing swelling and preventing ion accumulation that could otherwise disrupt microstructural order [50].

To further investigate the intrinsic properties of ONRs, conventional OECT devices (cOECTs) were constructed for geometry-independent evaluation with pivotal performance parameters summarized in Table S7. Although the distinct architectures of cOECTs and vOECTs give rise to fundamentally different

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electrochemical doping pathways, which modulate ion injection dynamics and volumetric charge carrier distribution, planar devices are still valuable for lateral comparisons of material performance. COECT devices based on **4NR** exhibited superior current amplification capability, with I_{on}/I_{off} of ~10^5 compared to ~10^4 for **2NR** and **3NR**. Meanwhile, the most pronounced enhancement is observed in $g_{m,\,max}$, which increases from 0.13 mS (**2NR**) and 0.18 mS (**3NR**) to 0.76 mS for **4NR**-based devices (Figure S29). After normalization by Wd/L, $g_{m,\,norm}$ was 0.18, 0.33, and 0.95 S cm^{-1} for **2NR**, **3NR**, and **4NR**, respectively, indicating that the performance improvement originates from intrinsic material properties rather than dimensional effects, consistent with observations in vOECTs.

Calculated by the equation $\mu C^* = \frac{g_m}{|V_{th}-V_{GS}|} \cdot \frac{L}{Wd}$ (where μ is the charge carrier mobility, C^* is the volumetric capacitance; V_{th} is the threshold voltage and V_{GS} is the gate voltage), the μC^*_{ave} value (the product of μ and C^* is used to assess the mixed conducting performance of cOECT devices) of **4NR** was obtained as 2.84 F V^{-1} cm^{-1} s^{-1}, greatly surpassing the values of 0.90 and 0.87 F V^{-1} cm^{-1} s^{-1} for **2NR** and **3NR**, respectively, and is comparable to those reported for polymeric mixed ionic-electronic conductors with similar backbone features [51]. To gain a clearer understanding of ion and electron transport, electrochemical impedance spectroscopy (EIS) measurements (Figure S31) were performed to determine C^* values, which were 156, 125, and 154 F cm^{-3} for **2NR**, **3NR**, and **4NR**, respectively. Notably, despite exhibiting the highest degree of crystallinity as revealed by GIWAXS, **2NR** shows a C^* comparable to that of **4NR**. This behavior can be rationalized by the presence of ion-accessible regions at crystalline domain boundaries, where mismatches in packing can generate localized free volume that accommodates ions [52]. The electron mobility μ was further estimated by the division of μC^* and C^*. With increasing conjugation length from **2NR** to **4NR**, the μ_e value increased progressively from 5.7×10^{-3} to 1.84×10^{-2} cm^2 V^{-1} s^{-1}. This enhancement can be attributed to extended π-conjugation, which promotes intramolecular electronic delocalization and reduces transport barriers. This observation further indicates that, although π–π stacking distance slightly increases with conjugation length and side-chain coherence is diminished, continuous charge-hopping pathways are largely preserved. In summary, the systematic progression in the performance metrics from **2NR** to **4NR** validates the designed structural modifications, where the partially relaxed molecular packing reduces long-range lamellar coherence while preserving intramolecular charge delocalization and facilitating ion penetration. This balance provides a plausible explanation for the simultaneous observation of high transconductance and efficient ionic-electronic coupling transport.

2.5 | Biocompatibility

To evaluate the biocompatibility of **4NR** for skin-contacting bioelectronic applications, cytotoxicity assays were performed using L929 cells as the representative cell model. Live/dead fluorescence staining was conducted on days 2, 4, and 8 to assess cell viability and proliferation on **4NR**-coated substrates. Glass slides without any polymer coating served as the control. As shown in Figure S32, L929 cells cultured on **4NR** film exhibited robust viability and proliferation over the entire testing period, with dense, uniformly distributed green fluorescence indicative of live cells and no observable signs of cytotoxicity or detachment. The cell morphology and growth dynamics were comparable to those observed on control substrates, highlighting the benign cellular interface and excellent cytocompatibility of **4NR** films under the tested conditions. Relative cell viability was summarized in Figure S33 after 1 and 5 days of exposure to extracts from **4NR** thin films prepared under cOECTs devices conditions, tested at six various concentrations ranging from 4% to 100%. After 5 days of incubation, the relative cell viability remained above 100% for most tested concentrations, indicating minimal cytotoxicity of **4NR** across a broad concentration range. These results established its suitability for long-term integration in wearable and implantable bioelectronic devices [53].

2.6 | Applications in Human Sweat Detection

Given their efficient ion-electron coupling and well-defined molecular architecture, the ONR derivatives serve as an ideal platform for bioelectronic sensing of physiologically relevant electrolytes. Interestingly, as the applied gate bias increased, the devices exhibited anti-ambipolar behavior. When operated in 10 mM PBS, all devices exhibited typical n-type behavior and excellent stability, with no current attenuation observed at gate voltage below 0.55 V (Figure 4b). However, when the gate bias exceeded 0.55 V, anti-ambipolar characteristics gradually emerged, identified as pronounced current attenuation resembling a Gaussian-like profile (Figure 5a). Concurrently, a significant reduction in the peak drain current was observed during the backward sweep, indicating a decline in operational stability under high gate bias (Figure 5b). Typically, the emergence of anti-ambipolar characteristics arises from a delicate balance among charge-carrier injection, ionic gating, and electrochemical doping/de-doping processes [54, 55]. However, in our system, excessive swelling of hydrated ions may further contribute to this phenomenon by inducing microstructural deformation within the channel. To validate this hypothesis, transfer characteristics were recorded using NaCl electrolytes of varying concentrations from 0.1 to 1000 mM, as shown in Figure 5c. A clear decrease in current was observed when the electrolyte concentration reached 10 mM. At higher electrolyte concentrations, faster ion transport and stronger ionic coupling promoted rapid saturation of ion doping at relatively low biases. Yet, excessive ion permeation under high voltages leads to severe swelling of the channel layer, disrupting film morphology. Conversely, at low electrolyte concentrations, limited ionic strength hinders effective doping, thereby suppressing the anti-ambipolar response. These trends were further corroborated by the transient response curves, which displayed distinct waveform variations with electrolyte concentration (Figure 5d).

Sodium and potassium, the principal extracellular and intracellular cations, are essential for regulating muscle contraction, maintaining cardiac rhythm, and controlling osmotic pressure [56]. Prolonged loss of these ions disrupts osmotic balance and impairs nerve signal transmission, leading to muscle cramps and, in extreme cases, life-threatening hyponatremia [56]. Such risks are particularly pronounced among athletes, outdoor

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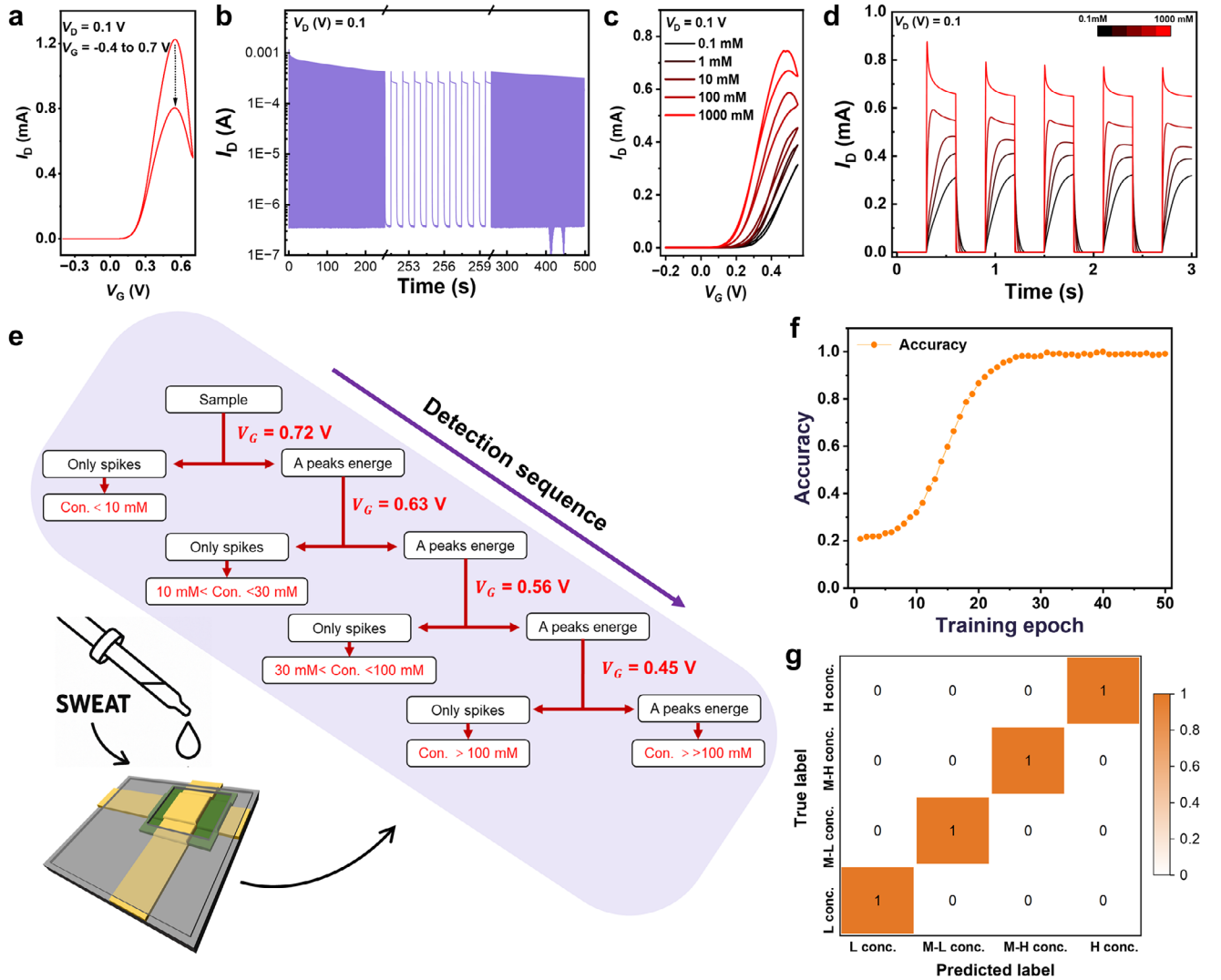


FIGURE 5 | Anti-ambipolar behavior and sweat concentration recognition based on machine learning. (a) Transfer curves of **4NR**-based OECT devices and operational stabilities over 500 cycles with alternating gate voltage from −0.4 to 0.7 V; (c) Transfer curves of **4NR**-based OECT device with alternating PBS electrolyte concentration ranging from 0.1 to 1000 mM; (d) Transient response curves of OECT device based on **4NR** in 0.1–1000 mM aqueous sodium chloride electrolyte; (e) Diagram of **4NR** based sweat detection progress; (f) Accuracy of the neural network model during 50 training epochs; (g) Cross-validated confusion matrix based on the anti-ambipolar recognition.

workers, and individuals in stressful occupations, where electrolyte imbalance can develop rapidly yet remain undetected until acute symptoms appear. The strong correlation between anti-ambipolar characteristics and ionic strength highlights the potential of **4NR**-based devices as sensitive platforms for electrolyte concentration monitoring.

The anti-ambipolar behavior is not interpreted as a failure mode in this work, but rather as a bias-dependent operating regime that can be harnessed for signal modulation under controlled conditions. By leveraging these anti-ambipolar characteristics manifested in the transfer curves and transient waveforms, a sensing model was developed to detect electrolyte levels in unknown sweat samples, enabling real-time health monitoring. As illustrated in Figure S34, transient response tests were conducted using proportionally diluted simulated body fluids (SBFs) with systematically varied Na^+ concentrations (10, 30, 100 mM), reflecting physiological ranges found in human sweats [57]. The transient waveforms obtained under different gate biases were categorized into two characteristic types: “spikes” and “A peaks” (anti-ambipolar peaks, featuring a current decay). This distinction can also be identified from the peak current trend, as the n-type “spike” mode maintains stable current levels, while the anti-ambipolar “A peak” mode exhibits reduced stability. As shown in Figure S30, transient curves displayed only “spikes” at a gate voltage of 0.45 V. “A peaks” first appeared at 100 mM Na^+ under a gate voltage of 0.56 V, were observed at both 30 mM and 100 mM SBF under 0.63 V, and became evident at all concentrations, including 10 mM, when the gate voltage increased to 0.72 V. These results demonstrate that the emergence threshold of anti-ambipolar behavior depends on Na^+ concentration, providing a reliable semi-quantitative indicator for estimating electrolyte levels in unknown sweat samples. A systematic testing sequence (from 0.72 to 0.45 V) was thus established to determine Na^+ concentration ranges, as summarized in the flowchart shown in Figure 5e.

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To enable intuitive and rapid estimation of analyte concentrations, machine learning was integrated into the sensing workflow, as outlined in the anti-ambipolar recognition flowchart shown in Figures S35 and S36. Artificial sweat samples with varying ionic concentrations were categorized into four levels: low (L conc., <10 mM Na^+), medium-low (M-L conc., 10–30 mM Na^+), medium-high (M-H conc., 30–100 mM Na^+), and high (H conc., >100 mM Na^+). For classification, waveform variations from the transient current responses were extracted as feature vectors and put into a random forest classifier. The mode was trained to learn the correlation between waveform characteristics and ionic concentration levels. The model exhibited an exceptionally high classification recognition rate. Specifically, the classification performance was quantitatively validated via recognition accuracy and confusion matrix analysis. Training results demonstrate that after 30 epochs, the model achieved a 100% recognition rate for distinct solution concentration levels. Notably, the confusion matrix consistently reflects the outstanding recognition precision, characterized by prominent diagonal dominance with negligible off-diagonal elements (Figure 5f–g), an observation that attests to the model's superior recognition accuracy and exceptional discriminative capability across all concentration categories. This anti-ambipolar behavior and classification performance were reproducibly maintained, with the recognition accuracy remaining largely unchanged, exhibiting only a minor decrease from 100% to 97% at the highest concentration level after 75 measurement cycles (Figure S37). These results demonstrated good operational robustness of the sensing platform under repeated measurements and controlled conditions.

This waveform-based sensing model offers significant advantages over conventional analytical approaches by eliminating the need for bulky instrumentation or time-consuming chemical assays. It enables on-site, non-invasive measurements with immediate feedback. Moreover, the stepwise classification based on transient waveform features and peak currents requires minimal computational resources, facilitating real-time data processing and seamless integration into wearable health-monitoring platforms.

3 | Conclusions

In summary, we developed a family of monodisperse π-extended ONR-based n-type OMIECs through a facile, metal-free step-economical π-extension strategy. This molecularly precise approach effectively eliminates batch-to-batch variation, enabling reproducible synthesis and accurate establishment of structure-property relationships. The resulting materials exhibit well-defined backbones, tunable crystallinity, and efficient ion-electron coupling, thereby addressing the long-standing challenges in achieving both high performance and stability in n-type monodisperse oligomeric OECTs. Structural analyses revealed that shorter conjugation lengths enhance crystallinity and π–π stacking, while moderate relaxation in extended systems promotes ion accessibility without disrupting electronic connectivity. This optimized structural balance enabled **4NR** to achieve great biocompatibility, exceptional operational stability, high transconductance, and rapid transient responses, outperforming most reported small-molecule counterparts. Moreover, leveraging the voltage-dependent, reconfigurable n-type and anti-ambipolar behaviors under physiological electrolytes, we established a waveform-based, non-invasive smart sweat-ion sensing platform. Coupled with a machine-learning classifier, the system achieved high accuracy in distinguishing sweat ionic concentrations across four levels, demonstrating the feasibility of precise, real-time electrolyte monitoring. Collectively, these results underscore the potential of monodisperse π-extended OMIECs as a versatile material platform for next-generation intelligent and wearable bioelectronic sensing technologies.

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Conflicts of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: admt71070-sup-0001-SuppMat.docx.

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